

Week 7: Polyphase Structures

Homework, Weekly Quiz, Complete Solutions, and Instructor Grading Notes

Course context: DSP + Machine Learning for a mixed senior / first-year ECE cohort. DSP reference: Oppenheim & Schaffer §4.7–4.8. ML bridge: structured convolution and fixed/learned subband front ends.

Important precision point: The Noble identities do not allow an arbitrary filter to move through a rate changer. The downsampling identity applies to $G(z^M)$, and the interpolation identity applies to $G(z^L)$.

Learning targets

- Decompose an FIR filter into polyphase components and reconstruct the original impulse response.
- Explain the arithmetic savings of polyphase decimation and interpolation.
- Apply the First and Second Noble identities with the correct z^M / z^L conditions.
- Describe a two-channel analysis/synthesis filter bank and perfect reconstruction.
- Compare polyphase time-phase partitioning with grouped/depthwise CNN computation and fixed subband feature extraction.

Homework — Student Version

Problem 1 — Polyphase decomposition (12 pts) [Core]

Let $h[n] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $M = 3$.

- Find $e_0[r]$, $e_1[r]$, and $e_2[r]$ using $e_k[r] = h[rM+k]$.
- Write $H(z)$ in polyphase form $H(z) = \sum_k z^{-k} E_k(z^M)$.
- Explain how the original tap ordering is recovered when the phases are recombined.

Problem 2 — Where the decimation savings comes from (10 pts) [Core]

A 48-tap FIR anti-aliasing filter precedes a downsampler by $M = 6$. The input rate is 96 kHz.

- Estimate the multiply rate for the naive filter-then-downsample implementation.
- Estimate the multiply rate for an ideal polyphase implementation.
- State the approximate speedup and explain why the number of multiplies per retained output is not reduced by the same factor.

Problem 3 — First Noble Identity and its limitation (12 pts) [Core + concept trap]

Let $G(z) = 1 + 2z^{-1}$ and $M = 2$. Use $x[n] = \delta[n]$.

- Compute the output of $G(z^2)$ followed by $\downarrow 2$.
- Compute the output of $\downarrow 2$ followed by $G(z)$. Verify equivalence.
- Now replace $G(z^2)$ in the left system by the arbitrary filter $H(z) = 1 + 2z^{-1}$. Show that $H(z) \rightarrow \downarrow 2$ is not equivalent to $\downarrow 2 \rightarrow H(z)$. Explain the lesson.

Problem 4 — Efficient decimator derivation (14 pts) [Senior derivation]

For $y[m]$ obtained by filtering $x[n]$ with $h[n]$ and then downsampling by M , start from $y[m] = \sum_q h[q]x[mM-q]$. Write $q=rM+k$ and derive a polyphase expression involving $e_k[r] = h[rM+k]$. What does the result say about which input sample phases are needed by each branch?

Problem 5 — Efficient interpolation phases (12 pts) [Core]

Let $h[n] = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and interpolation factor $L = 4$.

- (a) Find the four polyphase impulse responses.
- (b) Which phase produces output samples $y[4m+2]$?
- (c) If the original FIR has $N = 40$ taps, compare the approximate multiply rates of naive zero-insertion + full-rate filtering and polyphase interpolation.

Problem 6 — Haar analysis/synthesis bank (14 pts) [Core / filter bank]

Use the pairwise Haar transform

$$a[m] = (x[2m] + x[2m+1])/\sqrt{2} \text{ and } d[m] = (x[2m] - x[2m+1])/\sqrt{2}.$$

For $x = \{3, 1, 2, -2\}$:

- (a) Compute $a[m]$ and $d[m]$.
- (b) Reconstruct x exactly using the synthesis equations.
- (c) Verify that input energy equals total subband energy.

Problem 7 — Perfect-reconstruction reasoning (12 pts) [Senior concept]

A two-channel critically sampled analysis/synthesis bank uses analysis filters $H_0(z)$, $H_1(z)$ and synthesis filters $F_0(z)$, $F_1(z)$.

- (a) In words, what two things must be true for perfect reconstruction?
- (b) Interpret the alias-cancellation condition $F_0(z)H_0(-z) + F_1(z)H_1(-z) = 0$.
- (c) Why is “the two filters cover low and high frequencies” not by itself sufficient to guarantee perfect reconstruction?

Problem 8 — DSP/ML architecture synthesis (14 pts) [DSP + ML synthesis]

You must classify short vibration signals on a low-power embedded processor. Compare two front ends:

- A. Four fixed FIR subbands followed by four log-energy features and a small classifier.
- B. A learned Conv1d front end using grouped/depthwise convolution followed by a pointwise layer.

For each option discuss computation, trainable parameters, interpretability, adaptability, and one failure mode. Then recommend a design for a small labeled dataset and justify your choice. Finally, explain precisely how grouped convolution is similar to—but not mathematically the same as—polyphase decomposition.

Homework — Complete Solutions

Solution 1 — Polyphase decomposition (12 pts) [Core]

Solution

$$e_0 = \{1, 4, 7, 10\}, \quad e_1 = \{2, 5, 8\}, \quad e_2 = \{3, 6, 9\}$$

$$E_0(z) = 1 + 4z^{-1} + 7z^{-2} + 10z^{-3}$$

$$E_1(z) = 2 + 5z^{-1} + 8z^{-2}, \quad E_2(z) = 3 + 6z^{-1} + 9z^{-2}$$

$$H(z) = E_0(z^3) + z^{-1}E_1(z^3) + z^{-2}E_2(z^3)$$

The factor z^{-k} restores the phase offset k . Expanding $E_k(z^3)$ places its coefficients every third delay, and the outer z^{-k} shifts them back to indices $k, k+3, k+6, \dots$.

Grading guidance:

- 4 pts phase sequences; 5 pts z -domain form; 3 pts interpretation.

Solution 2 — Decimation savings (10 pts) [Core]

Solution

$$\text{naive} \approx 48 \cdot 96,000 = 4.608 \times 10^6 \text{ multiplies/s}$$

$$\text{polyphase} \approx (48/6) \cdot 96,000 = 0.768 \times 10^6 \text{ multiplies/s}$$

The ideal speedup is about 6×. Each retained output still depends on roughly all 48 FIR taps; the savings comes from computing only 1/6 as many output instants, not from using only 1/6 of the taps in a retained convolution sum.

Grading guidance:

- 3 pts naive rate; 3 pts polyphase rate; 4 pts explanation.

Solution 3 — First Noble Identity (12 pts) [Core]

Solution

(a) $G(z^2) = 1 + 2z^{-2}$, so with $\delta[n]$ the filter output is $\{1, 0, 2\}$; $\downarrow 2$ retains $\{1, 2\}$.

(b) $\delta[n]$ downsampled by 2 is still $\delta[m]$. Filtering by $G(z) = 1 + 2z^{-1}$ gives $\{1, 2\}$. The systems match.

(c) For $H(z) = 1 + 2z^{-1}$, $H(z)\delta = \{1, 2\}$; $\downarrow 2$ retains only $\{1\}$. But $\downarrow 2\delta = \delta$ followed by H gives $\{1, 2\}$. They do not match. The Noble Identity requires the special $G(z^M)$ spacing; an arbitrary filter cannot simply be moved through $\downarrow M$.

Grading guidance:

- 3 pts part a; 3 pts part b; 4 pts counterexample; 2 pts lesson.

Solution 4 — Efficient decimator derivation (14 pts) [Core]

Solution

$$\begin{aligned}y[m] &= \sum_q h[q] x[mM-q] \\q=rM+k \Rightarrow y[m] &= \sum_{k=0}^{M-1} \sum_r h[rM+k] x[mM-rM-k] \\y[m] &= \sum_k \sum_r e_k[r] x[(m-r)M-k]\end{aligned}$$

For branch k , the input samples have indices congruent to $-k$ modulo M (with the exact apparent phase depending on the chosen delay/commutator drawing). Each E_k filter can therefore operate on a phase-selected low-rate sequence rather than on the full-rate stream. Summing the branch outputs reproduces exactly the retained full-rate convolution samples.

Grading guidance:

- 5 pts index substitution; 5 pts final expression; 4 pts phase/rate interpretation.

Solution 5 — Efficient interpolation phases (12 pts) [Core]

Solution

$$e_0=\{1,5\}, \quad e_1=\{2,6\}, \quad e_2=\{3,7\}, \quad e_3=\{4,8\}$$

Phase e_2 produces the outputs $y[4m+2]$. In general, $y[mL+k]=(e_k*x)[m]$ for the usual causal phase ordering.

For $N=40$ and $L=4$, naive full-rate FIR work is approximately $40 \cdot 4 \cdot F_s = 160F_s$ multiplies/s. Polyphase work is approximately $40F_s$ multiplies/s total, an ideal $4\times$ saving.

Grading guidance:

- 4 pts phases; 3 pts output phase; 5 pts operation-rate comparison.

Solution 6 — Haar filter bank (14 pts) [Core]

Solution

$$\begin{aligned}a[0] &= (3+1)/\sqrt{2} = 2\sqrt{2}, \quad d[0] = (3-1)/\sqrt{2} = \sqrt{2} \\a[1] &= (2-2)/\sqrt{2} = 0, \quad d[1] = (2-(-2))/\sqrt{2} = 2\sqrt{2}\end{aligned}$$

Reconstruction: $x[0]=(2\sqrt{2}+\sqrt{2})/\sqrt{2}=3$, $x[1]=(2\sqrt{2}-\sqrt{2})/\sqrt{2}=1$, $x[2]=(0+2\sqrt{2})/\sqrt{2}=2$, $x[3]=(0-2\sqrt{2})/\sqrt{2}=-2$.

$$\begin{aligned}\text{input energy} &= 3^2+1^2+2^2+(-2)^2 = 18 \\ \text{subband energy} &= (2\sqrt{2})^2+(\sqrt{2})^2+0^2+(2\sqrt{2})^2 = 8+2+0+8 = 18\end{aligned}$$

The orthonormal Haar transform is energy preserving and perfectly reconstructing.

Grading guidance:

- 5 pts analysis coefficients; 5 pts reconstruction; 4 pts energy check.

Solution 7 — Perfect reconstruction (12 pts) [Core]

Solution

- Aliasing created by the downsamplers must cancel when the synthesis branches are added.
- The desired non-aliased transfer path must reduce to only a constant scale and pure delay: $\hat{x}[n]=c \cdot x[n-d]$.
- The condition $F_0(z)H_0(-z)+F_1(z)H_1(-z)=0$ says that the folded/aliased contribution from one branch is exactly canceled by the folded contribution from the other.
- Frequency coverage alone is insufficient because the filters also have phase, transition-band, amplitude, and synthesis relationships. Two filters can split low/high energy yet still leave aliasing or amplitude/phase distortion after reconstruction.

Grading guidance:

- 4 pts two requirements; 4 pts alias-condition interpretation; 4 pts why coverage is insufficient.

Solution 8 — DSP/ML architecture synthesis (14 pts) [Core]

Solution

- Fixed four-band front end: very few or no trainable front-end parameters, predictable computation, interpretable band energies, and good data efficiency. Failure mode: the chosen bands may discard task-relevant structure or be mismatched to a new operating condition.
- Grouped/depthwise Conv1d: fewer parameters/MACs than dense convolution and can adapt filters to the task; a pointwise layer restores cross-channel mixing. Failure modes include learned aliasing when stride is used, overfitting with limited labels, or learned filters that are difficult to interpret.
- For a small labeled dataset on a tight embedded budget, a fixed DSP bank plus tiny classifier is a strong baseline. A hybrid—DSP initialization followed by limited fine-tuning—can be tested if enough validation data exists.
- Similarity: both exploit structured independent branches and later recombine results. Difference: polyphase partitions taps/samples by time phase modulo the rate-change factor and is an exact reorganization of a multirate FIR. Grouped convolution partitions feature channels and is a learnable architectural constraint; it is not a Noble-identity/polyphase transformation.

Grading guidance:

- 4 pts fixed front end; 4 pts grouped/learned front end; 3 pts recommendation; 3 pts precise analogy/difference.

Weekly Quiz — Student Version

Suggested time: 15–20 minutes. Total: 20 points. Basic calculator allowed.

1. (2 pts) For $M=3$, write the definition of the k -th polyphase sequence $e_k[r]$ in terms of $h[n]$.

2. (2 pts) A 30-tap FIR is followed by $\downarrow 5$. Approximately what ideal arithmetic speedup does a polyphase decimator provide relative to filtering every full-rate output?

3. (3 pts) True or false: $H(z) \rightarrow \downarrow M$ is always equivalent to $\downarrow M \rightarrow H(z)$. Explain.

4. (3 pts) For $h=\{1,2,3,4,5,6\}$ and $M=2$, find e_0 and e_1 .

5. (2 pts) State the Second Noble Identity in words or block-diagram form.

6. (3 pts) A 24-tap FIR interpolates by $L=3$. Compare the approximate naive and polyphase multiply rates in units of F_s .

7. (2 pts) In a two-channel perfect-reconstruction bank, what two broad requirements must be satisfied?

8. (3 pts) Give one similarity and one important difference between polyphase decomposition and grouped convolution.

Weekly Quiz — Solutions and Grading

1. Polyphase definition

$e_k[r] = h[rM+k]$, $k=0, \dots, M-1$.

Grading: 2 pts.

2. Speedup

Approximately $5\times$.

Grading: 2 pts; accept explanation that only one of five output instants survives.

3. Noble-identity trap

False. A filter may commute across $\downarrow M$ in the Noble identity when it has the special $G(z^M)$ form before the downsampler; arbitrary $H(z)$ does not commute.

Grading: 1 pt false; 2 pts condition/explanation.

4. $M=2$ decomposition

$e_0 = \{1, 3, 5\}$, $e_1 = \{2, 4, 6\}$.

Grading: 1.5 pts each.

5. Second Noble Identity

$\uparrow L$ followed by $G(z^L)$ is equivalent to $G(z)$ followed by $\uparrow L$.

Grading: 2 pts.

6. Interpolation work

Naive $\approx 24 \cdot 3F_s = 72F_s$ multiplies/s; polyphase $\approx 24F_s$; about $3\times$ saving.

Grading: 1 pt each rate; 1 pt speedup.

7. Perfect reconstruction

Alias terms cancel, and the desired transfer path is only a scale and delay (no frequency-dependent distortion).

Grading: 1 pt each.

8. Polyphase vs grouped conv

Similarity: both partition work into parallel structured branches. Difference: polyphase partitions time/tap phases and exactly reorganizes a multirate FIR; grouped convolution partitions learned feature channels.

Grading: 1.5 pts similarity; 1.5 pts difference.

Instructor Notes and Differentiation

For first-year students

- Use tap-index coloring and explicit $M=2$ or $M=3$ sequences before z -domain notation.
- Grade the Noble identities primarily as “special spaced filter can move to the low-rate side,” rather than requiring a formal proof.
- Use the Haar pairwise average/difference example as the main filter-bank anchor.

For senior students

- Require the $q=rM+k$ decimator derivation and careful input-phase indexing.
- Require the alias-cancellation / distortion-transfer equations for two-channel perfect reconstruction.
- Extension: derive a polyphase matrix representation for the analysis bank or implement a custom decimator and compare timing against `scipy.signal.upfirdn`.